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Lecture 14: Dynamic Optimal Transport

1 The Riemannian Structure of the 2–Wasserstein Distance: Gradient Flows and Linearization

The lecture mainly talks about the probability measures with the 2-Wasserstein metric, its dynamic formulation, the formal Riemannian structure, and two key applications: gradient flows of energy functionals leading to PDEs and linearization (LOT) useful in learning.

1.1 The Monge problem

Definition 1 (The Monge Problem). *Let $\Omega \subset \mathbb{R}^n$ be a bounded, convex domain. Denote by $P(\Omega)$ the set of probability measures on Ω and $\rho_0, \rho_1 \in P(\Omega)$. We seek for a measurable map $T\colon \Omega \to \Omega$ transport ρ_0 onto ρ_1 that satisfies:*

$$\int_\Omega f(x)\, d\rho_1(x) = \int_\Omega f(T(x)\, d\rho_0(x) \quad \forall f \in L^\infty(\Omega)$$

In this case,

$$W_2^2(\rho_0, \rho_1) = \inf_T \left\{ \int_\Omega \|T(x) - x\|^2 \, d\rho_0(x) \ : \ T_\# \rho_0 = \rho_1 \right\}.$$

1.2 Wasserstein Geodesics

Definition 2 (Wasserstein Geodesics). *Given T as above, define the Wasserstein geodesics:*

$$T_t = (1-t)\,\mathrm{Id} + t\,T, \qquad \rho_t = (T_t)_\# \rho_0, \qquad t \in [0,1].$$

Then $\{\rho_t\}$ is a constant-speed geodesic in $(P(\Omega), W_2)$:

$$W_2(\rho_t, \rho_s) = |t-s|\, W_2(\rho_0, \rho_1), \qquad \forall s, t \in [0,1].$$

Contrast with linear interpolation of measures: the convex mixture $\tilde{\rho}_t = (1-t)\rho_0 + t\rho_1$ is not a Wasserstein geodesic and does not solve the continuity equation below.

1.3 Formal Riemannian Structure

$M \in P(\Omega)$ behaves like an infinite-dimensional Riemannian manifold where we set up the following:

- Tangent space $T_\rho M$ at each point $\rho \in M$
- Inner Product $\langle \cdot, \cdot \rangle_\rho$ on $T_\rho M$
- Corresponding norm $||w||_\rho^2 = \langle w, w \rangle_\rho$ induces W_2 distance:

$$W_2^2(\rho_0, \rho_1) = \inf_{\rho_t} \left\{ \int_0^1 ||\partial_t \rho_t||_{\rho_t}^2 \, dt \colon \rho_t|_{t=0} = \rho_0, \rho_t|_{t=1} = \rho_1 \right\}$$

The above is an infinite dimensional problem. This structure degenerates as we go from smooth, strictly positive measure ($d\mu(x) = f(x)dx, f > 0, f \in C^\infty$) to $P(\Omega)$. For

$$T_\rho M = \{\partial_t \rho_t|_{t=0} \ : \ \rho_t \text{ is a smooth curve in } M \text{ with } \rho_t|_{t=0} = \rho_0\},$$

the above induced W_2 distance is based on Benamou-Brenier Theorem and the fact that if we are given vectors h_i at ρ, we can find ψ_i just by solving the generalized Poisson equation which allows us to invert the Laplacian:

$$\langle h_1, h_2 \rangle_\rho = \int \nabla\psi_1 \cdot \nabla\psi_2 \, d\rho$$
$$h_i = -\nabla \cdot (\nabla\psi_i \rho)$$

We have that

$$\begin{aligned}
W_2^2(\rho_0, \rho_1) &= \inf_{(\rho_t, v_t)} \left\{ \int_0^1 \int_\Omega |v_t|^2 d\rho_t \, dt \ : \ \partial\rho_t + \nabla \cdot (v_t \rho_t) = 0, \ \rho_t|_{t=0} = \rho_0, \ \rho_t|_{t=1} = \rho_1 \right\} \\
&= \inf_{\rho_t} \left\{ \inf_{v_t : \partial\rho_t = -\nabla\cdot(v_t\rho_t)} \int_0^1 \int_\Omega |v_t|^2 d\rho_t \, dt \ : \ \rho_t|_{t=0} = \rho_0, \ \rho_t|_{t=1} = \rho_1 \right\} \\
&= \inf_{\rho_t} \left\{ \int_0^1 \int_\Omega |\nabla\psi_t|^2 d\rho_t \, dt \ : \ \rho_t|_{t=0} = \rho_0, \ \rho_t|_{t=1} = \rho_1, \ \partial\rho_t = -\nabla \cdot (\psi_t \rho_t) \right\} \\
&= \inf_{\rho_t} \left\{ \int_0^1 \|\partial_t \rho\|_{\rho_t}^2 \, dt \ : \ \rho_t|_{t=0} = \rho_0, \ \rho_t|_{t=1} = \rho_1, \right\}
\end{aligned}$$

1.4 Dynamic characterization of W_2

For any smooth time-dependent curve ρ_t in $P(\Omega)$, there exists a velocity field v_t such that it solves the continuity equation:

$$\begin{cases} \partial_t \rho_t + \nabla \cdot (v_t \rho_t) = 0 \\ \rho_t|_{t=0} = \rho_0 \end{cases}$$

All the paths are solutions to the continuity equation. *Example*: If $\rho_t = T_t \# \rho_0$, one can show that $v_t \circ T_t = T_{\rho_0}^{\rho_t} - I$, so for all $t \in [0, 1]$

$$W_2^2(\rho_0, \rho_1) = \inf_\Omega |T_{\rho_0}^{\rho_1} - I|^2 \, d\rho_0 = \int_\Omega |v_t \circ T_t|^2 \, d\rho_0 = \int_\Omega |v_t|^2 \, d\rho_t.$$

This can be reformulated into the following theorem:

Theorem 1. *[BB00].*

$$W_2^2(\rho_0, \rho_1) = \inf_{(\rho_t, v_t)} \left\{ \int_0^1 \int_\Omega |v_t|^2 \, d\rho_t \, dt \ : \ \partial_t \rho_t + \nabla \cdot (v_t \rho_t) = 0, \ \rho_{|t=0} = \rho_0, \ \rho_{|t=1} = \rho_1 \right\}.$$

Note:

- If (ρ_t, v_t) satisfies the continuity equation, then $(\rho_t v_t + w_t)$ also solves for $\nabla \cdot (w_t \rho_t) = 0$.
- $||v + w||_{L^2(\mu)} \geq ||v||_{L^2(\mu)}$, $\forall w$ s.t. $\nabla \cdot (w\mu) = 0$ if and only if

$$v \in \overline{\{\nabla\psi \ : \ \psi \in C_c^\infty(\mathbb{R}^d)\}}^{L^2(\mu)},$$

  i.e if v is in the $L^2(\mu)$ closure of gradients of smooth functions of $\mathbb{R}^d$.

1.5 Gradient flows for W_2

Let E be a continuous functional with first variation $\frac{\delta E}{\delta \rho}$. The gradient flow with respect to metric W_2 of E solves

$$\begin{aligned}
\langle \nabla_{W^2} E(\rho_0), \partial_t \rho_t|_{t=0} \rangle_{\rho_0} &= \lim_{h \to 0} \frac{E(\rho_t) - E(\rho_0)}{h} \\
&= \int \frac{\partial E}{\partial \rho_0} \cdot \partial_t \rho_t|_{t=0} \, dx \\
&= \int \frac{\partial E}{\partial \rho_0} \cdot (-\nabla \cdot (\nabla \psi_0 \rho_0)) \, dx \\
&= \int \nabla \frac{\delta E}{\delta \rho_0} \cdot \psi_0 \, d\rho_0 \\
&= \left\langle -\nabla \cdot \left(\nabla \frac{\partial E}{\partial \rho_0} \rho_0 \right), \partial_t \rho_t|_{t=0} \right\rangle_{\rho_0}
\end{aligned}$$

Example 1. *By properly choosing the functional we can recover "classical" differential equations.*

1. Fokker–Planck: *For* $E(\rho) = \int (V\rho + \rho \log \rho) \, dx$*, one gets* $\partial_t \rho = \nabla \cdot (\rho \nabla V) + \Delta \rho$.
2. Aggregation + drift:, $E(\rho) = \frac{1}{2} \int \int K(x-y) \, d\rho(x) \, d\rho(y) \, dx + \int V \, d\rho$*, one gets* $\partial_t \rho = \nabla \cdot (\rho \nabla (K * \rho)) + \nabla \cdot (\rho \nabla V)$.
3. Nonlinear diffusion equations: $E(\rho) = \int f(\rho) + \int V\rho$*, one gets* $\partial_t \rho = \nabla \cdot (\rho \nabla f(\rho)) + \nabla \cdot (\rho \nabla V)$.

1.6 Linearized 2-Wasserstein Metric

Recall: If $\rho_t = T_t \# \rho_0$, then $v_t \circ T_t = T_{\rho_0}^{\rho_1} - I \Rightarrow v_0 = T_{\rho_0}^{\rho_1} - I = \nabla \psi$. In this case, $||\partial_t \rho_t|_{t=0}||_{\rho_0}^2 = \int |\nabla \psi|^2 \, d\rho_0 = \int |T_{\rho_0}^{\rho_1} - I|^2 \, d\rho_0 = W_2^2(\rho_0, \rho_1)$. Thus, to linearize W_2 about ρ_0:

$$W_2^2(\rho_1, \tilde{\rho}_1) \approx ||\partial_t \rho_t|_{t=0} - \partial_t \tilde{\rho}_t|_{t=0}||_{\rho_0}^2 = \int |(T_{\rho_0}^{\rho_1} - I) - (T_{\rho_0}^{\tilde{\rho}_1} - I)| = \int |T_{\rho_0}^{\rho_1} - T_{\rho_0}^{\tilde{\rho}_1}|^2 \, d\rho_0 = \mathrm{LOT}_{\rho_0}(\rho_1, \tilde{\rho}_1).$$

Benefits of LOT:

- LOT_{ρ_0} is a metric. Also, $W_2(\mu, \nu) \leq \mathrm{LOT}_{\mu_0}(\mu, \nu) \leq C W_2(\mu, \nu)^{1/6}$. [MDC20], [DM23], [LM24].
- Computing LOT_{ρ_0} between $O(N)$ images requires $O(N)$ OT calculations and $O(N^2) L^2(\rho_0)$ calculations.
- Embedding into linear space $\mu \longrightarrow T_{\rho_0}^{\mu} - I$.

Applications

- Jet Tagging: The goal is to use HCAL measurements to infer what occur at parton level.
- Event Classification with LOT [CCCC20].

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